

Playfair Cipher – Encryption

10. Write a C program for Playfair matrix:

M F H I/J K

U N O P Q

Z V W X Y

E L A R G

D S T B C

Encrypt this message: Must see you over Cadogan West. Coming at once.

Aim

To write a C program to encrypt the given message using the supplied **Playfair matrix**.

Given Matrix

M F H I/J K

U N O P Q

Z V W X Y

E L A R G

D S T B C

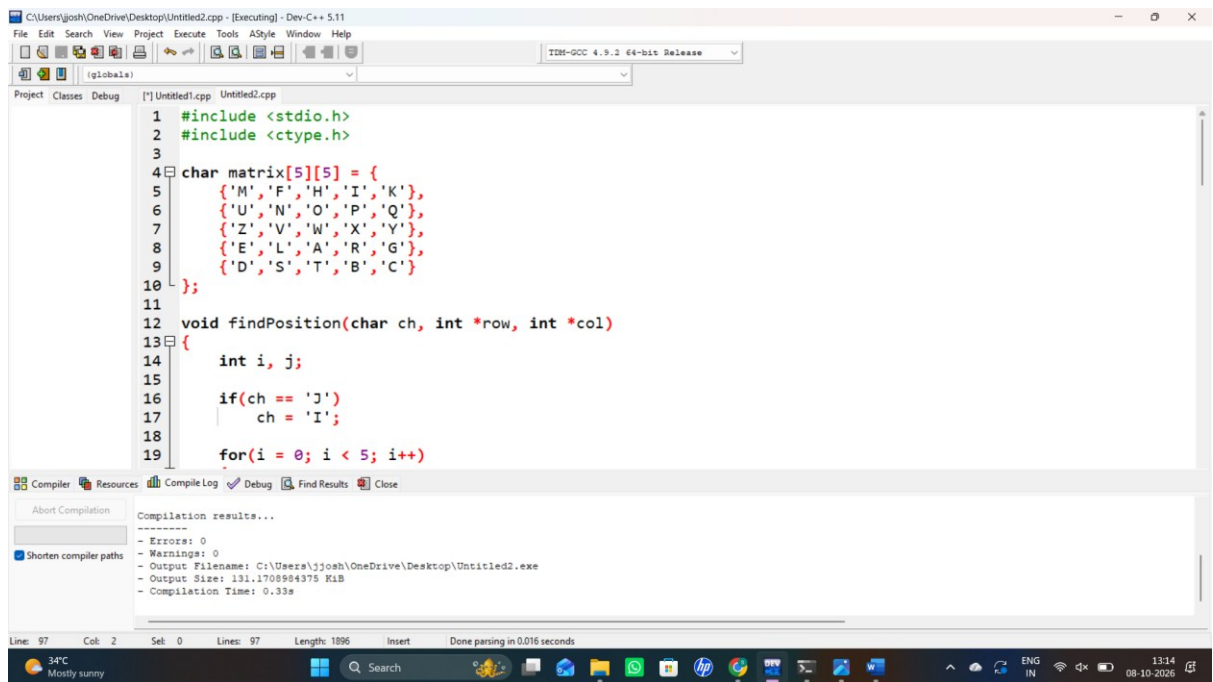
Plaintext

MUST SEE YOU OVER CADOGAN WEST. COMING AT ONCE.

Algorithm

1. Construct the given 5×5 Playfair matrix.
2. Convert plaintext to uppercase.
3. Remove spaces and punctuation.
4. Replace J with I.
5. Divide plaintext into pairs.
6. If both letters are in the same row, move right.
7. If both letters are in the same column, move down.
8. Otherwise, form a rectangle and exchange columns.
9. Display ciphertext

Program

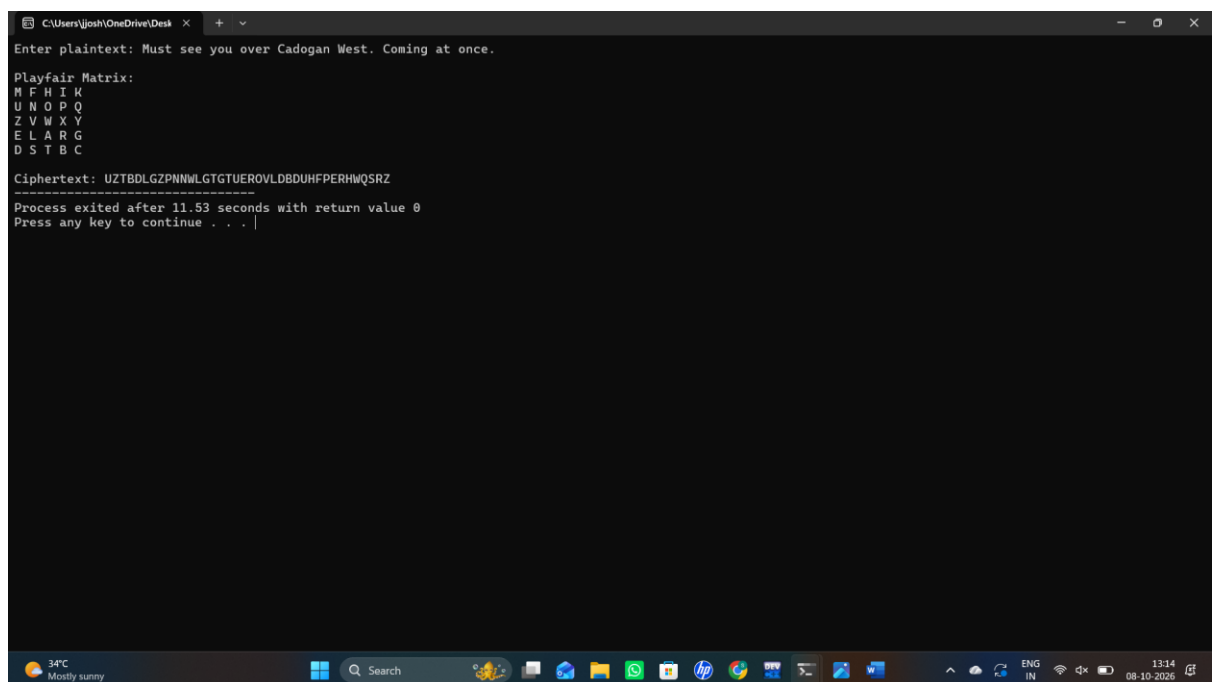


```
1 #include <stdio.h>
2 #include <ctype.h>
3
4 char matrix[5][5] = {
5     {'M','F','H','I','K'},
6     {'U','N','O','P','Q'},
7     {'Z','V','W','X','Y'},
8     {'E','L','A','R','G'},
9     {'D','S','T','B','C'}
10 };
11
12 void findPosition(char ch, int *row, int *col)
13 {
14     int i, j;
15
16     if(ch == 'J')
17         ch = 'I';
18
19     for(i = 0; i < 5; i++)
```

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\josh\OneDrive\Desktop\Untitled2.exe
- Output Size: 131.1708994375 KiB
- Compilation Time: 0.33s

Input/Output



```
Enter plaintext: Must see you over Cadogan West. Coming at once.

Playfair Matrix:
M F H I K
U N O P Q
Z V W X Y
E L A R G
D S T B C

Ciphertext: UZTBDLGZPNNWLGTTUEROVLDBDUHFPERHWQSRZ

Process exited after 11.53 seconds with return value 0
Press any key to continue . . .
```

Result

Thus, the given message was successfully encrypted using the **Playfair cipher**.